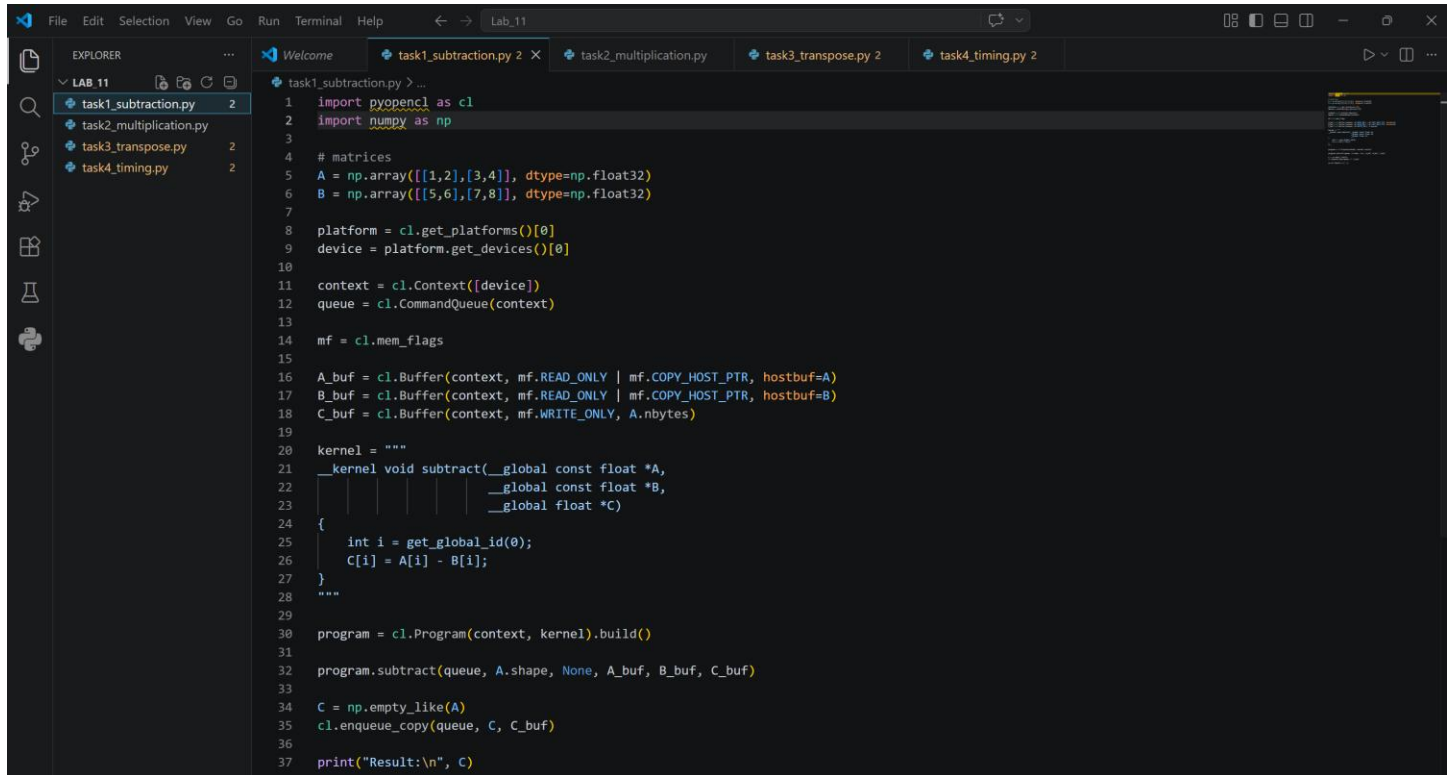
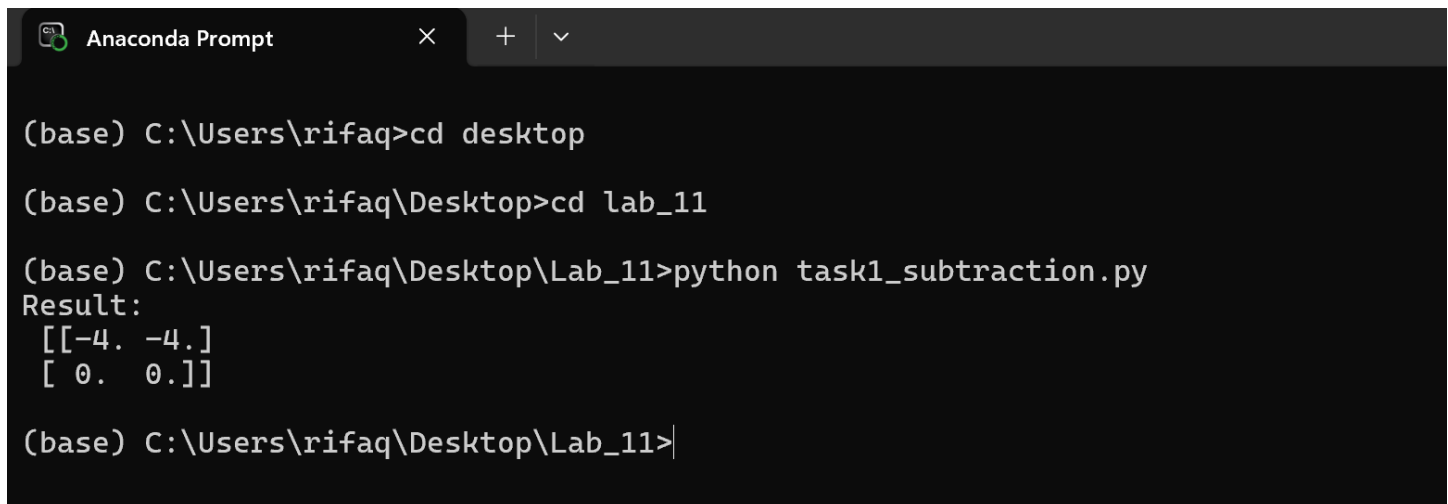


TASK : 1



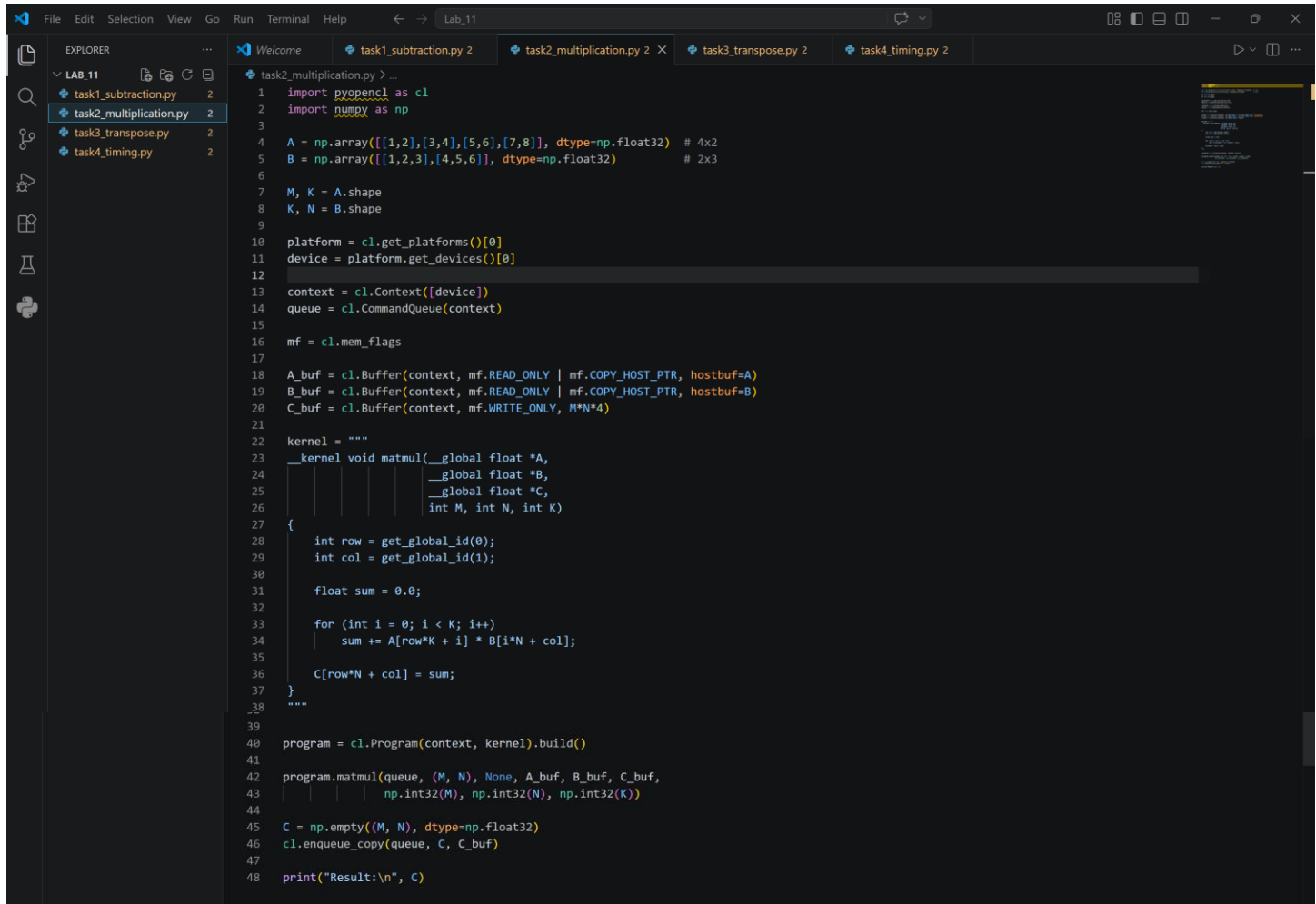
```
1 import pyopencl as cl
2 import numpy as np
3
4 # matrices
5 A = np.array([[1,2],[3,4]], dtype=np.float32)
6 B = np.array([[5,6],[7,8]], dtype=np.float32)
7
8 platform = cl.get_platforms()[0]
9 device = platform.get_devices()[0]
10
11 context = cl.Context([device])
12 queue = cl.CommandQueue(context)
13
14 mf = cl.mem_flags
15
16 A_buf = cl.Buffer(context, mf.READ_ONLY | mf.COPY_HOST_PTR, hostbuf=A)
17 B_buf = cl.Buffer(context, mf.READ_ONLY | mf.COPY_HOST_PTR, hostbuf=B)
18 C_buf = cl.Buffer(context, mf.WRITE_ONLY, A.nbytes)
19
20 kernel = """
21 _kernel void subtract(__global const float *A,
22                     __global const float *B,
23                     __global float *C)
24 {
25     int i = get_global_id(0);
26     C[i] = A[i] - B[i];
27 }
28 """
29
30 program = cl.Program(context, kernel).build()
31
32 program.subtract(queue, A.shape, None, A_buf, B_buf, C_buf)
33
34 C = np.empty_like(A)
35 cl.enqueue_copy(queue, C, C_buf)
36
37 print("Result:\n", C)
```

OUTPUT



```
(base) C:\Users\rifaq>cd desktop
(base) C:\Users\rifaq\Desktop>cd lab_11
(base) C:\Users\rifaq\Desktop\Lab_11>python task1_subtraction.py
Result:
[[-4. -4.]
 [ 0.  0.]]
(base) C:\Users\rifaq\Desktop\Lab_11>
```

TASK : 2



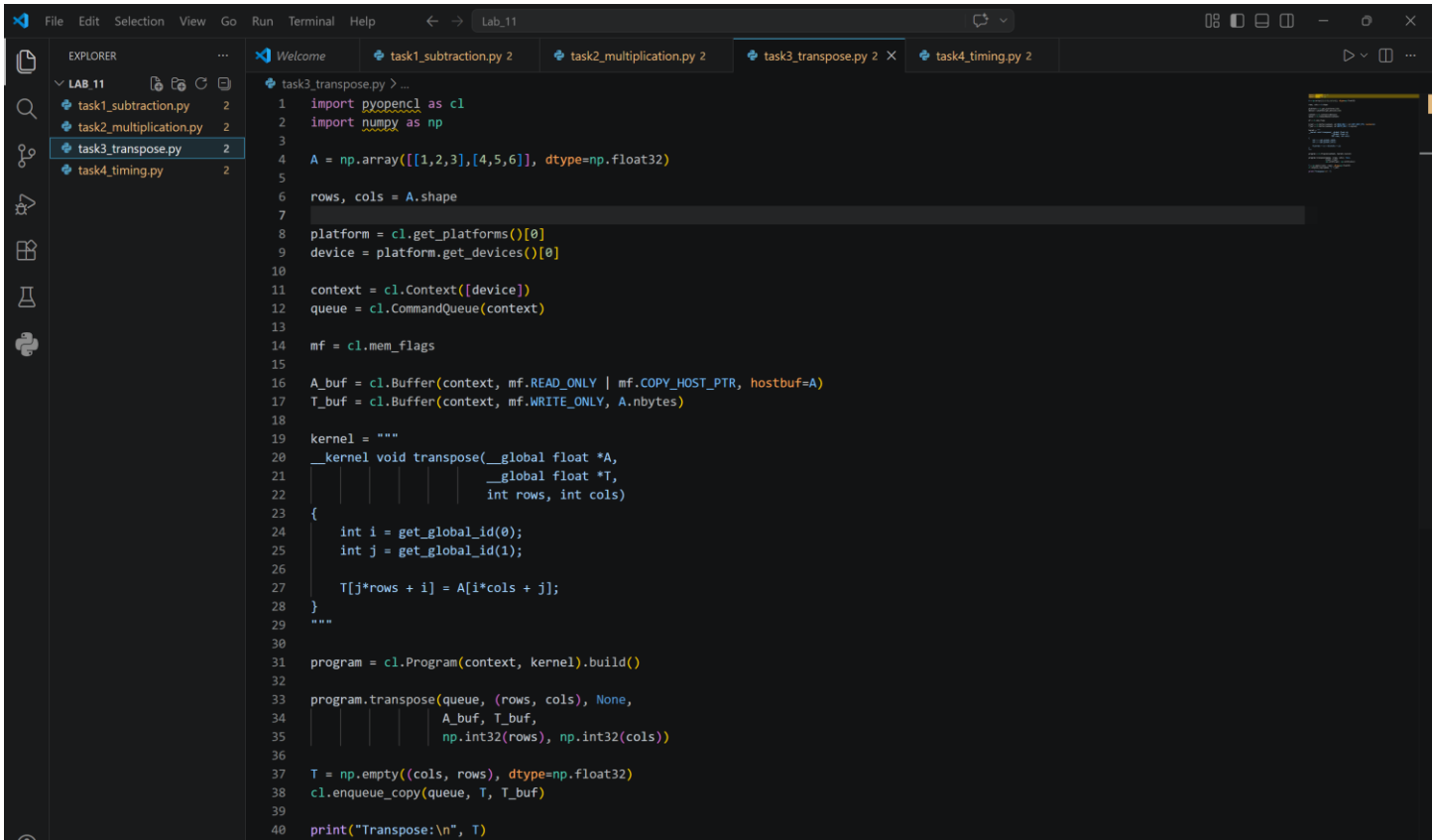
```
1 import pyopencl as cl
2 import numpy as np
3
4 A = np.array([[1,2],[3,4],[5,6],[7,8]], dtype=np.float32) # 4x2
5 B = np.array([[1,2,3],[4,5,6]], dtype=np.float32) # 2x3
6
7 M, K = A.shape
8 K, N = B.shape
9
10 platform = cl.get_platforms()[0]
11 device = platform.get_devices()[0]
12
13 context = cl.Context([device])
14 queue = cl.CommandQueue(context)
15
16 mf = cl.mem_flags
17
18 A_buf = cl.Buffer(context, mf.READ_ONLY | mf.COPY_HOST_PTR, hostbuf=A)
19 B_buf = cl.Buffer(context, mf.READ_ONLY | mf.COPY_HOST_PTR, hostbuf=B)
20 C_buf = cl.Buffer(context, mf.WRITE_ONLY, M*N*4)
21
22 kernel = """
23 _kernel void matmul(__global float *A,
24                   __global float *B,
25                   __global float *C,
26                   int M, int N, int K)
27 {
28     int row = get_global_id(0);
29     int col = get_global_id(1);
30
31     float sum = 0.0;
32
33     for (int i = 0; i < K; i++)
34         sum += A[row*K + i] * B[i*N + col];
35
36     C[row*N + col] = sum;
37 }
38 """
39
40 program = cl.Program(context, kernel).build()
41
42 program.matmul(queue, (M, N), None, A_buf, B_buf, C_buf,
43               np.int32(M), np.int32(N), np.int32(K))
44
45 C = np.empty((M, N), dtype=np.float32)
46 cl.enqueue_copy(queue, C, C_buf)
47
48 print("Result:\n", C)
```

OUTPUT

```
(base) C:\Users\rifaq\Desktop\Lab_11>python task2_multiplication.py
Result:
[[ 9. 12. 15.]
 [19. 26. 33.]
 [29. 40. 51.]
 [39. 54. 69.]]

(base) C:\Users\rifaq\Desktop\Lab_11>
```

TASK : 3



```
1 import pyopencl as cl
2 import numpy as np
3
4 A = np.array([[1,2,3],[4,5,6]], dtype=np.float32)
5
6 rows, cols = A.shape
7
8 platform = cl.get_platforms()[0]
9 device = platform.get_devices()[0]
10
11 context = cl.Context([device])
12 queue = cl.CommandQueue(context)
13
14 mf = cl.mem_flags
15
16 A_buf = cl.Buffer(context, mf.READ_ONLY | mf.COPY_HOST_PTR, hostbuf=A)
17 T_buf = cl.Buffer(context, mf.WRITE_ONLY, A.nbytes)
18
19 kernel = """
20 _kernel void transpose(__global float *A,
21                       __global float *T,
22                       int rows, int cols)
23 {
24     int i = get_global_id(0);
25     int j = get_global_id(1);
26
27     T[j*rows + i] = A[i*cols + j];
28 }
29 """
30
31 program = cl.Program(context, kernel).build()
32
33 program.transpose(queue, (rows, cols), None,
34                  A_buf, T_buf,
35                  np.int32(rows), np.int32(cols))
36
37 T = np.empty((cols, rows), dtype=np.float32)
38 cl.enqueue_copy(queue, T, T_buf)
39
40 print("Transpose:\n", T)
```

OUTPUT

```
(base) C:\Users\rifaq\Desktop\Lab_11>python task3_transpose.py
Transpose:
[[1. 4.]
 [2. 5.]
 [3. 6.]]

(base) C:\Users\rifaq\Desktop\Lab_11>
```

TASK : 4**OUTPUT**

```
(base) C:\Users\rifaq\Desktop\Lab_11>python task4_timing.py
CPU Time: 0.002238750457763672
GPU(OpenCL) Time: 0.05684328079223633
```

```
(base) C:\Users\rifaq\Desktop\Lab_11>
```